

1. Administrative & Study Identification

Full Study Title: Open-Label Rater-Blind Randomized Multi-Center Parallel-Arm Active- Comparator Study to Assess the Efficacy and Tolerability of Ofatumumab 20mg SC Monthly vs. First Line DMT - Physician's Choice in the Treatment of Newly Diagnosed RMS **Short Title/Acronym:** Open Label Randomized Multicenter to Assess Efficacy & Tolerability of Ofatumumab 20mg vs. First Line DMT in RMS (STHENOS) **Posting ID:** 451588077401067684 **Protocol Number:** EudraCT: 2020-004505-32 **NCT Number:** NCT04788615 **Sponsor Name:** Novartis Pharmaceuticals (Novartis Pharma AG) **Trial Status:** Active, not recruiting (ACTIVE_NOT_RECRUITING) **Investigational Product:** Ofatumumab 20mg (Trade names: Arzerra, Kesimpta | ATC Codes: L01FA02, L04AA52) **Phase of Development:** Phase 3 **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To provide clinical, MRI, and patient-reported outcome (PRO) data demonstrating a better efficacy, patient quality of life, and compliance of ofatumumab in a treatment-naïve Relapsing Multiple Sclerosis (RMS) population compared to the current standard of care (SoC) first-line disease-modifying treatments (DMT). **Secondary Objectives:** To evaluate the number of relapses, annual relapse rate, percentage of relapse-free patients, and proportion of relapse-free patients with MRI activity free, alongside evaluations of disability worsening, brain volume changes, and serum neurofilament light chain levels.

2.2 Background & Rationale To address the gap in early interventional MS treatments by providing real-world evidence. This study is being conducted to support local patient access processes and to establish ofatumumab 20mg SC monthly as an early, highly-effective first-line intervention for newly diagnosed RMS patients compared to traditional physician's choice therapies.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (First Line DMT - physician's choice) **Blinding:** Open-label, Rater-blind **Randomization:** Randomized (1:1 Parallel-arm)

3.2 Planned Enrollment & Duration Target **\$N\$:** 185 Targeted Enrollment (planned for 186 total patients) **Number of Sites:** Multicenter (Multiple centers across Europe, including UK and Spain) **Estimated Study Start:** 23-Jul-2021 **Estimated Primary Completion:** 04-Nov-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male or female patients, 18 through 55 (inclusive) years of age. 2. Diagnosis of Multiple Sclerosis according to the 2017 revised McDonald criteria. 3. Relapsing MS: relapsing-course (RMS). 4. Treatment-naïve patients (≤ 5 years since first MS symptom) with an EDSS score of 0-4.0 (inclusive). 5. At least 1 relapse or 1 Gd+ enhanced lesion on T1 in the 1 year prior to Screening. 6. Patient must be suitable to be treated with one of first line self-administered DMT-physician's choice.

4.2 Exclusion Criteria 1. Progressive MS phenotypes: Patients with primary progressive MS (PPMS) or Secondary Progressive MS (SPMS). 2. Positive results at Screening for serological markers for Hepatitis B (HBV) and C (HCV) indicating acute or chronic infection, or HIV/COVID-19 active infections. 3. Patients with neurological findings consistent with Progressive Multifocal Leukoencephalopathy (PML), or confirmed PML. 4. Pregnant or nursing women, or women of childbearing potential not using highly effective contraception. 5. Use of other experimental or investigational drugs at the time of enrollment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Ofatumumab 20mg monthly vs. First-line DMT (physician's choice) **Product Details:** Ofatumumab (Trade names: Arzerra, Kesimpta | ATC Codes: L01FA02, L04AA52) **EMA Information:** [Arzerra EPAR](#), [Kesimpta EPAR](#) **Route of Administration:** Subcutaneous (SC) injection for ofatumumab **Duration of Treatment:** Up to 15 months maximum duration.

5.2 Concomitant & Prohibited Therapy Permitted: Standard concomitant medications for symptomatic MS management that do not interact with the study DMTs. **Prohibited:** Concurrent use of alternative MS disease-modifying therapies (DMTs) outside the assigned study treatment arms or experimental drugs.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -14 to 0 | Informed Consent, Medical History, Hepatitis B/C serology, Baseline MRI | | **Baseline** | Day 0 | Randomization, First Dose of Ofatumumab or

Comparator | | **Interim** | Month 3 | Safety Labs, Clinical Efficacy Assessment, MRI re-baselining | | **End of Study** | Month 15 | Final Efficacy/Disability Assessment, Exit Interview, Safety follow-up (6-month observational extension safety period for those who withdraw early) |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of participants with no evidence of disease activity (NEDA-3). Defined as: 1. Absence of confirmed clinical relapse 2. Absence of new MRI activity (Gd+ T1 lesion or new/enlarged T2 lesion) with MRI re-baselined at Month 3 3. Absence of 3-month confirmed disability worsening

Measurement Method: Clinical relapse assessments, EDSS scoring, and rater-blinded Central MRI Reading.

7.2 Secondary & Exploratory Endpoints 1. Number of relapses. 2. Annual relapse rate. 3. Percentage of relapse-free patients and proportion of relapse-free patients with MRI activity free. 4. (Exploratory) Disability worsening at 3 months or 6 months, and disability improvement at 6 months; annualized rate of new/enlarging lesions on T2-weighted MRI and serum neurofilament light chain levels.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection at all visits; specialized monitoring for injection-related reactions and Hepatitis B/C reactivation risk. **Serious Adverse Events (SAEs):**

Reported within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Internal/External independent safety monitoring board setup per Novartis standards.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) Set for efficacy and Safety Set for AEs. **Sample Size Justification:** $N = 185$

appropriately powered ($\alpha = 0.05$, $Power = 80\%$) to detect a statistically significant difference in NEDA-3 achievement.

Handling of Missing Data: Multiple Imputation (MI) or Mixed-Effects Models for Repeated Measures (MMRM) for continuous secondary endpoints.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard onsite/remote monitoring ensuring

proper blinding of independent MRI raters. **Audit Trail:** Robust eCRF system utilizing data clarification forms.

11. Study Identifiers & Governance

Primary ID: 2020-004505-32 **Registry Number:** NCT04788615 **Posting ID:** 451588077401067684 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** STHENOS **Official Regulatory Title:** Open-Label Rater-Blind Randomized Multi-Center Parallel-Arm Active- Comparator Study to Assess the Efficacy and Tolerability of Ofatumumab 20mg SC Monthly vs. First Line DMT - Physician's Choice in the Treatment of Newly Diagnosed RMS

12. Clinical Framework (Multiple Sclerosis Specific)

Primary Condition: Multiple Sclerosis **Preferred UMLS Name:** Relapsing-Remitting Multiple Sclerosis **Diagnosis Threshold:** 2017 revised McDonald criteria; Treatment Naïve (≤ 5 years) **Medical Methodology:** First-Line Disease Modifying Treatment (DMT) **Optimization Target:** Achievement of No Evidence of Disease Activity (NEDA-3)

13. Study Procedures & Methodology

Management Pathway: Standard induction of DMT vs. Ofatumumab targeted therapy. **Education Protocol (HCP):** Blinded-rater protocol training for unbiased EDSS/MRI clinical evaluations. **Education Protocol (Patient):** Subcutaneous auto-injection technique training for Ofatumumab. **Quality Audit Mechanism:** Independent blinded MRI reading centers to ensure objective primary endpoint calculations.

14. Observation & Follow-Up Schedule

Total Duration: 15 months (plus a 6-month safety observational extension for patients discontinuing early for any reason) **Onsite Visit Cadence:** Baseline, Month 3, and periodic visits up to Month 15. **Remote Monitoring Frequency:** Between scheduled clinic visits (eCRF tracking). **Checklist Review Interval:** End of treatment (EOT) tracking and final reconciliation.

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	____	[DD-MMM-YYYY]				